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EDUCATION

RV College of Engineering

B-Tech in Computer Science and Engineering

Bangalore, India

CGPA: 9.7

Expected Graduation 2027

SKILLS

Technologies/ Frameworks: LLMs, Transformers, Hugging Face Transformers, Scikit-learn, TensorFlow, PyTorch, Flask, Express.js, Node.js, React, Flutter, React Native, Firebase, Firebase Cloud Messaging (FCM), Linux

Languages: C, C++, Java, Python, R, JavaScript, Dart, PHP, HTML/CSS

Databases: MongoDB, MySQL, Hive

Libraries: NumPy, Pandas, SciPy, NLTK, SpaCy, D3.js, Plotly, Google Gemini API Client, Azure SDK

EXPERIENCE

Project Lead – AI-based Telecom Fault Detection System

March. – July. 2025

Project Lead · RVCE IBM Department

Bangalore

- Led cross-functional coordination, research, and planning for a collaborative AI project under the IBM Department at RVCE.
- Spearheaded integration of deep learning models (CNN-BiLSTM-Attention) with OTDR trace analysis to support intelligent fault detection in optical networks.
- Guided team members in pipeline design, model experimentation, and evaluation strategies; ensured delivery milestones were met.
- Secured 2nd place at the IBM Internal Project Showcase, among department-wide AI research presentations.

PROJECTS

ExamFlow – Blockchain-Powered Assessment Platform

- Engineered a full-stack examination system with real-time AI-based proctoring, integrity scoring, and tamper-proof certification using Ethereum (ERC-721) and IPFS.
- Developed adaptive mock test generation with dynamic difficulty, performance analytics, and personalized preparation insights.
- Designed modular dashboards for students and instructors with live monitoring, behavioral logs, and on-chain certificate verification.

SentinelDDoS – Advanced DDoS Detection and Mitigation System

- Engineered a modular DDoS protection system using entropy-based algorithms, statistical analysis, and ML models for real-time anomaly detection and mitigation.
- Built a traffic profiling engine and entropy calculator to establish behavioral baselines and detect anomalies using Shannon entropy, Z-scores, and adaptive thresholds.
- Designed automated mitigation strategies including rate-limiting, IP blocking, and behavioral filtering with policy enforcement via Python-based middleware.
- Integrated a real-time attack simulator with customizable parameters to assess and fine-tune defense mechanisms against varied attack vectors.

AI-Powered Fault Detection in Optical Networks

- Designed a deep learning pipeline (CNN + BiLSTM + Attention) to detect and localize faults in OTDR traces, achieving ~97.5% accuracy even in low SNR environments.
- Built a preprocessing pipeline with Savitzky-Golay filtering, normalization, and segmentation to prepare raw OTDR signals for accurate model input.
- Designed automated mitigation strategies including rate-limiting, IP blocking, and behavioral filtering with policy enforcement via Python-based middleware.
- Implemented a React.js-based interface and REST API for OTDR trace uploads, visualization, and near real-time fault analysis.

ACTIVITIES

- Achieved runners-up in the IBMxRVCE Project showcase for developing an AI-powered OTDR Fault Localization System under the IOT Department at RVCE.
- Secured Top 5 position nationally at Hackfest (Hackatjo), a 36-hour on-site blockchain hackathon hosted by NMAM Institute of Technology, competing among 60+ top engineering college teams across India.
- Contributed technical summaries for AI and Signal Processing research papers during project development and implementation.
- Practiced and solved 200+ coding problems across LeetCode and GeeksforGeeks